# DAY 4

In [2]:
```
import pandas as pd
```

In [4]:
```
df=pd.read_csv("world-happiness-report - world-happiness-report.csv")
```

In [5]:
```
df.head()
```

Out[5]:

| | Country name | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | Generosity | Perce corr |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | 50.80 | 0.718 | 0.168 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | 51.20 | 0.679 | 0.190 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | 51.60 | 0.600 | 0.121 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | 51.92 | 0.496 | 0.162 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | 52.24 | 0.531 | 0.236 | |

In [6]:
```
df.tail()
```

Out[6]:

| | Country name | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | Generosity | Per co |
|---|---|---|---|---|---|---|---|---|---|
| **1944** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | 54.4 | 0.733 | -0.095 | |
| **1945** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | 55.0 | 0.753 | -0.098 | |
| **1946** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | 55.6 | 0.763 | -0.068 | |
| **1947** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | 56.2 | 0.632 | -0.064 | |
| **1948** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | 56.8 | 0.643 | -0.009 | |

In [7]:
```
df.shape
```

Out[7]:
```
(1949, 11)
```

In [8]:
```
df.columns
```

Out[8]:
```
Index(['Country name', 'year', 'Life Ladder', 'Log GDP per capita',
       'Social support', 'Healthy life expectancy at birth',
       'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Positive affect', 'Negative affect'],
      dtype='object')
```

In [9]:
```
df
```

Out[9]:

| | Country name | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | Generosity | Pe c |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | 50.80 | 0.718 | 0.168 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | 51.20 | 0.679 | 0.190 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | 51.60 | 0.600 | 0.121 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | 51.92 | 0.496 | 0.162 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | 52.24 | 0.531 | 0.236 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **1944** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | 54.40 | 0.733 | -0.095 | |
| **1945** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | 55.00 | 0.753 | -0.098 | |
| **1946** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | 55.60 | 0.763 | -0.068 | |
| **1947** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | 56.20 | 0.632 | -0.064 | |
| **1948** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | 56.80 | 0.643 | -0.009 | |

1949 rows × 11 columns

In [10]:
```
df.columns
```

Out[10]:
```
Index(['Country name', 'year', 'Life Ladder', 'Log GDP per capita',
       'Social support', 'Healthy life expectancy at birth',
       'Freedom to make life choices', 'Generosity',
       'Perceptions of corruption', 'Positive affect', 'Negative affect'],
      dtype='object')
```

In [11]:
```
df.sample(8)
```

Out[11]:

| | Country name | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | Generosity | Pe co |
|---|---|---|---|---|---|---|---|---|---|
| **982** | Lesotho | 2011 | 4.898 | 7.777 | 0.824 | 45.74 | 0.618 | -0.087 | |
| **873** | Jordan | 2019 | 4.453 | 9.201 | 0.793 | 67.00 | 0.726 | -0.165 | |
| **1844** | United States | 2014 | 7.151 | 10.956 | 0.902 | 68.62 | 0.866 | 0.221 | |
| **476** | Dominican Republic | 2014 | 5.387 | 9.582 | 0.891 | 64.68 | 0.905 | -0.020 | |
| **133** | Belarus | 2006 | 5.658 | 9.489 | 0.918 | 61.10 | 0.707 | -0.246 | |
| **1642** | Sweden | 2011 | 7.382 | 10.814 | 0.921 | 71.98 | 0.941 | 0.161 | |
| **1583** | South Korea | 2008 | 5.390 | 10.383 | 0.754 | 70.80 | 0.524 | -0.102 | |
| **431** | Cyprus | 2015 | 5.439 | 10.445 | 0.770 | 73.10 | 0.628 | 0.114 | |

In [12]:

```
df.info
```

Out[12]:
```
<bound method DataFrame.info of      Country name  year  Life Ladder  Log GDP p
er capita  Social support  \
0     Afghanistan  2008        3.724                7.370           0.451
1     Afghanistan  2009        4.402                7.540           0.552
2     Afghanistan  2010        4.758                7.647           0.539
3     Afghanistan  2011        3.832                7.620           0.521
4     Afghanistan  2012        3.783                7.705           0.521
...           ...   ...          ...                  ...             ...
1944     Zimbabwe  2016        3.735                7.984           0.768
1945     Zimbabwe  2017        3.638                8.016           0.754
1946     Zimbabwe  2018        3.616                8.049           0.775
1947     Zimbabwe  2019        2.694                7.950           0.759
1948     Zimbabwe  2020        3.160                7.829           0.717

      Healthy life expectancy at birth  Freedom to make life choices  \
0                                50.80                         0.718
1                                51.20                         0.679
2                                51.60                         0.600
3                                51.92                         0.496
4                                52.24                         0.531
...                                ...                           ...
1944                             54.40                         0.733
1945                             55.00                         0.753
1946                             55.60                         0.763
1947                             56.20                         0.632
1948                             56.80                         0.643

      Generosity  Perceptions of corruption  Positive affect  Negative affect
0          0.168                      0.882            0.518            0.258
1          0.190                      0.850            0.584            0.237
2          0.121                      0.707            0.618            0.275
3          0.162                      0.731            0.611            0.267
4          0.236                      0.776            0.710            0.268
...          ...                        ...              ...              ...
1944      -0.095                      0.724            0.738            0.209
1945      -0.098                      0.751            0.806            0.224
1946      -0.068                      0.844            0.710            0.212
1947      -0.064                      0.831            0.716            0.235
1948      -0.009                      0.789            0.703            0.346

[1949 rows x 11 columns]>
```

In [13]:
```
df.dtypes
```

Out[13]:
```
Country name                         object
year                                  int64
Life Ladder                         float64
Log GDP per capita                  float64
Social support                      float64
Healthy life expectancy at birth    float64
Freedom to make life choices        float64
Generosity                          float64
Perceptions of corruption           float64
Positive affect                     float64
Negative affect                     float64
dtype: object
```

In [14]:
```
df.describe()
```

Out[14]:

| | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | |
|---|---|---|---|---|---|---|---|
| **count** | 1949.000000 | 1949.000000 | 1913.000000 | 1936.000000 | 1894.000000 | 1917.000000 | 1 |
| **mean** | 2013.216008 | 5.466705 | 9.368453 | 0.812552 | 63.359374 | 0.742558 | |
| **std** | 4.166828 | 1.115711 | 1.154084 | 0.118482 | 7.510245 | 0.142093 | |
| **min** | 2005.000000 | 2.375000 | 6.635000 | 0.290000 | 32.300000 | 0.258000 | |
| **25%** | 2010.000000 | 4.640000 | 8.464000 | 0.749750 | 58.685000 | 0.647000 | |
| **50%** | 2013.000000 | 5.386000 | 9.460000 | 0.835500 | 65.200000 | 0.763000 | |
| **75%** | 2017.000000 | 6.283000 | 10.353000 | 0.905000 | 68.590000 | 0.856000 | |
| **max** | 2020.000000 | 8.019000 | 11.648000 | 0.987000 | 77.100000 | 0.985000 | |

In [15]:
```
df.isnull()
```

Out[15]:

| | Country name | year | Life Ladder | Log GDP per capita | Social support | Healthy life expectancy at birth | Freedom to make life choices | Generosity | Perce corr |
|---|---|---|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False | False | False | |
| **1** | False | False | False | False | False | False | False | False | |
| **2** | False | False | False | False | False | False | False | False | |
| **3** | False | False | False | False | False | False | False | False | |
| **4** | False | False | False | False | False | False | False | False | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | |
| **1944** | False | False | False | False | False | False | False | False | |
| **1945** | False | False | False | False | False | False | False | False | |
| **1946** | False | False | False | False | False | False | False | False | |
| **1947** | False | False | False | False | False | False | False | False | |
| **1948** | False | False | False | False | False | False | False | False | |

1949 rows × 11 columns

In [16]:
```
df.isnull().sum()
```

Out[16]:
```
Country name                        0
year                                0
Life Ladder                         0
Log GDP per capita                 36
Social support                     13
Healthy life expectancy at birth   55
Freedom to make life choices       32
Generosity                         89
Perceptions of corruption         110
Positive affect                    22
Negative affect                    16
dtype: int64
```

In [17]:
```
df.columns=df.columns.str.strip().str.lower().str.replace(" ","_")
df.columns
```

Out[17]:
```
Index(['country_name', 'year', 'life_ladder', 'log_gdp_per_capita',
       'social_support', 'healthy_life_expectancy_at_birth',
       'freedom_to_make_life_choices', 'generosity',
       'perceptions_of_corruption', 'positive_affect', 'negative_affect'],
      dtype='object')
```

In [18]:
```
df.isnull().sum()
```

Out[18]:
```
country_name                        0
year                                0
life_ladder                         0
log_gdp_per_capita                 36
social_support                     13
healthy_life_expectancy_at_birth   55
freedom_to_make_life_choices       32
generosity                         89
perceptions_of_corruption         110
positive_affect                    22
negative_affect                    16
dtype: int64
```

In [19]:
```
important_cols = [
    'life_ladder',
    'log_gdp_per_capita',
    'social_support',
    'healthy_life_expectancy_at_birth',
    'freedom_to_make_life_choices'
]

for col in important_cols:
    df[col] = df[col].fillna(df[col].mean())
```

In [20]:
```
df['generosity'] = df['generosity'].fillna(df['generosity'].mean())
df['perceptions_of_corruption'] = df['perceptions_of_corruption'].fillna(df['per
```

In [21]:
```
df=df.dropna()
```

In [22]:
```
df.isnull().sum()
```

Out[22]:
```
country_name                       0
year                               0
life_ladder                        0
log_gdp_per_capita                 0
social_support                     0
healthy_life_expectancy_at_birth   0
freedom_to_make_life_choices       0
generosity                         0
perceptions_of_corruption          0
positive_affect                    0
negative_affect                    0
dtype: int64
```

In [23]:
```
df.shape
```

Out[23]:
```
(1925, 11)
```

In [24]:
```
df = df.drop_duplicates()
```

In [42]:
```
df
```

Out[42]:

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_( |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1944** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1945** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1946** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1947** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1948** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 11 columns

In [25]:
```
df.groupby('country_name')['life_ladder'].mean()
```

Out[25]:
```
country_name
Afghanistan    3.594667
Albania        5.019385
Algeria        5.379286
Angola         4.420250
Argentina      6.310133
                 ...
Venezuela      6.019867
Vietnam        5.315923
Yemen          3.912250
Zambia         4.551714
Zimbabwe       3.882600
Name: life_ladder, Length: 164, dtype: float64
```

In [26]:
```
df.groupby('country_name').size()
```

Out[26]:
```
country_name
Afghanistan    12
Albania        13
Algeria         7
Angola          4
Argentina      15
               ..
Venezuela      15
Vietnam        13
Yemen          12
Zambia         14
Zimbabwe       15
Length: 164, dtype: int64
```

In [27]:
```
df.groupby('country_name')['life_ladder'].agg(['mean', 'min', 'max'])
```

Out[27]:

| | mean | min | max |
|---|---|---|---|
| **country_name** | | | |
| **Afghanistan** | 3.594667 | 2.375 | 4.758 |
| **Albania** | 5.019385 | 4.511 | 5.867 |
| **Algeria** | 5.379286 | 4.745 | 6.355 |
| **Angola** | 4.420250 | 3.795 | 5.589 |
| **Argentina** | 6.310133 | 5.793 | 6.776 |
| **...** | ... | ... | ... |
| **Venezuela** | 6.019867 | 4.041 | 7.478 |
| **Vietnam** | 5.315923 | 5.023 | 5.767 |
| **Yemen** | 3.912250 | 2.983 | 4.809 |
| **Zambia** | 4.551714 | 3.307 | 5.260 |
| **Zimbabwe** | 3.882600 | 2.694 | 4.955 |

164 rows × 3 columns

In [28]:
```
df.groupby(['country_name', 'year'])['life_ladder'].mean().reset_index()
```

Out[28]:

| | country_name | year | life_ladder |
|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 |
| **1** | Afghanistan | 2009 | 4.402 |
| **2** | Afghanistan | 2010 | 4.758 |
| **3** | Afghanistan | 2011 | 3.832 |
| **4** | Afghanistan | 2012 | 3.783 |
| **...** | ... | ... | ... |
| **1920** | Zimbabwe | 2016 | 3.735 |
| **1921** | Zimbabwe | 2017 | 3.638 |
| **1922** | Zimbabwe | 2018 | 3.616 |
| **1923** | Zimbabwe | 2019 | 2.694 |
| **1924** | Zimbabwe | 2020 | 3.160 |

1925 rows × 3 columns

In [29]:
```
df.groupby('country_name')['life_ladder'].mean().reset_index()
```

Out[29]:

| | country_name | life_ladder |
|---|---|---|
| **0** | Afghanistan | 3.594667 |
| **1** | Albania | 5.019385 |
| **2** | Algeria | 5.379286 |
| **3** | Angola | 4.420250 |
| **4** | Argentina | 6.310133 |
| **...** | ... | ... |
| **159** | Venezuela | 6.019867 |
| **160** | Vietnam | 5.315923 |
| **161** | Yemen | 3.912250 |
| **162** | Zambia | 4.551714 |
| **163** | Zimbabwe | 3.882600 |

164 rows × 2 columns

In [30]:
```
df.groupby('year')['life_ladder'].mean().reset_index()
```

Out[30]:

| | year | life_ladder |
|---|---|---|
| **0** | 2005 | 6.564200 |
| **1** | 2006 | 5.202886 |
| **2** | 2007 | 5.430772 |
| **3** | 2008 | 5.417139 |
| **4** | 2009 | 5.461496 |
| **5** | 2010 | 5.499175 |
| **6** | 2011 | 5.414228 |
| **7** | 2012 | 5.443754 |
| **8** | 2013 | 5.390437 |
| **9** | 2014 | 5.375085 |
| **10** | 2015 | 5.397197 |
| **11** | 2016 | 5.399479 |
| **12** | 2017 | 5.462363 |
| **13** | 2018 | 5.507743 |
| **14** | 2019 | 5.578685 |
| **15** | 2020 | 5.859181 |

In [31]:
```
df.groupby('country_name')[['log_gdp_per_capita','life_ladder']].mean()
```

Out[31]:

| | log_gdp_per_capita | life_ladder |
|---|---|---|
| **country_name** | | |
| **Afghanistan** | 7.650833 | 3.594667 |
| **Albania** | 9.384385 | 5.019385 |
| **Algeria** | 9.334857 | 5.379286 |
| **Angola** | 8.990000 | 4.420250 |
| **Argentina** | 10.033800 | 6.310133 |
| **...** | ... | ... |
| **Venezuela** | 9.457757 | 6.019867 |
| **Vietnam** | 8.638308 | 5.315923 |
| **Yemen** | 8.313659 | 3.912250 |
| **Zambia** | 8.066857 | 4.551714 |
| **Zimbabwe** | 7.850333 | 3.882600 |

164 rows × 2 columns

In [32]:
```
df
```

Out[32]:

| | country_name | year | life_ladder | log_gdp_per_capita | social_support | healthy_life_ |
|---|---|---|---|---|---|---|
| **0** | Afghanistan | 2008 | 3.724 | 7.370 | 0.451 | |
| **1** | Afghanistan | 2009 | 4.402 | 7.540 | 0.552 | |
| **2** | Afghanistan | 2010 | 4.758 | 7.647 | 0.539 | |
| **3** | Afghanistan | 2011 | 3.832 | 7.620 | 0.521 | |
| **4** | Afghanistan | 2012 | 3.783 | 7.705 | 0.521 | |
| **...** | ... | ... | ... | ... | ... | |
| **1944** | Zimbabwe | 2016 | 3.735 | 7.984 | 0.768 | |
| **1945** | Zimbabwe | 2017 | 3.638 | 8.016 | 0.754 | |
| **1946** | Zimbabwe | 2018 | 3.616 | 8.049 | 0.775 | |
| **1947** | Zimbabwe | 2019 | 2.694 | 7.950 | 0.759 | |
| **1948** | Zimbabwe | 2020 | 3.160 | 7.829 | 0.717 | |

1925 rows × 11 columns

In [33]:
```
df.isnull().sum()
```

Out[33]:
```
country_name                        0
year                                0
life_ladder                         0
log_gdp_per_capita                  0
social_support                      0
healthy_life_expectancy_at_birth    0
freedom_to_make_life_choices        0
generosity                          0
perceptions_of_corruption           0
positive_affect                     0
negative_affect                     0
dtype: int64
```

In [34]:
```
df.to_csv("cleaned file.csv")
```

In [ ]: